

GazeAgent: From Gaze Estimation to Query-Driven Analysis in Naturalistic Interactions

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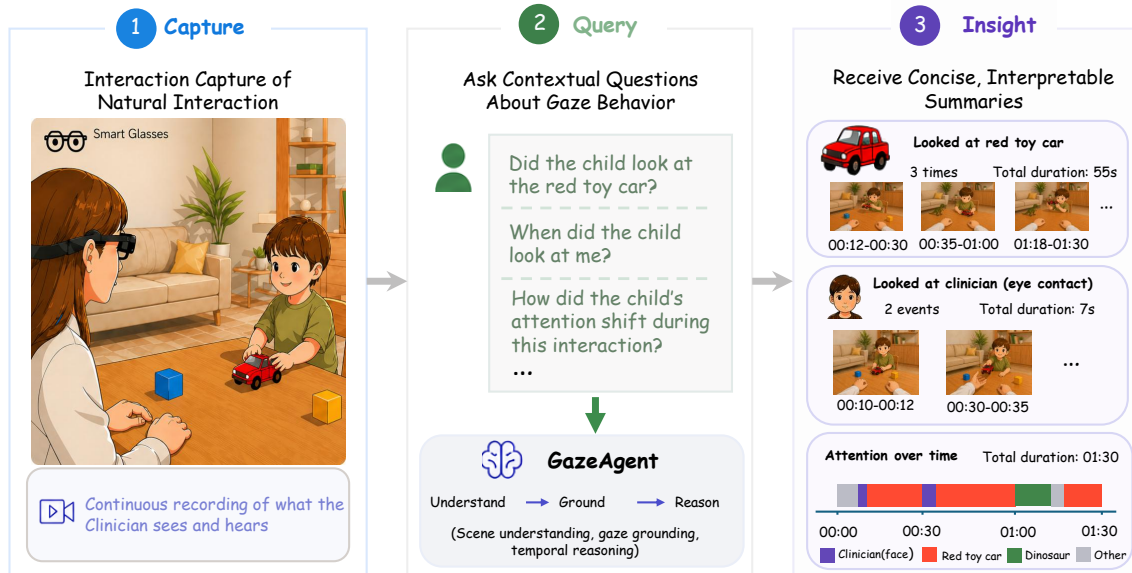


Fig. 1. System workflow of *GazeAgent*. **(1) Capture:** Unobtrusive egocentric recording of natural clinical interactions via smart glasses. **(2) Query:** Clinicians pose contextual, natural-language questions regarding specific gaze behaviors (e.g., object fixation or eye contact). **(3) Insight:** Powered by robust scene understanding, gaze grounding, and temporal reasoning, the agent translates these queries into concise analytical summaries and continuous attention timelines. (Part of the figure is created by Generative AI)

Understanding gaze behavior is central to behavioral assessment and developmental research, yet existing approaches remain limited by either intrusive sensing setups or labor-intensive post-hoc analysis workflows. While recent advances in gaze estimation improve low-level prediction accuracy, they do not directly support how practitioners analyze gaze in practice, which typically involves semantic and temporal questions such as attention targets, durations, and gaze shifts. We present **GazeAgent**, a system for query-driven analysis of gaze behavior in naturalistic interactions. GazeAgent reframes gaze analysis as a structured reasoning problem by transforming raw interaction data into a semantically grounded and temporally aligned representation. The system integrates gaze estimation with scene-aware grounding to construct a structured gaze timeline, and employs a code-generating agent to translate natural-language queries into executable analysis over this representation. This design enables precise computation of gaze statistics and temporal relationships while maintaining interpretability and reducing reliance on end-to-end generation. We evaluate GazeAgent through both technical benchmarks and user studies. Results demonstrate robust end-to-end reasoning performance across diverse query types. In user studies, GazeAgent significantly reduces analysis workload and improves task efficiency and user confidence in interpreting gaze behavior. Our work introduces a query-driven paradigm for gaze analysis that bridges perception and gaze-based behavioral analysis, enabling flexible, on-demand access to structured gaze insights in real-world interaction settings.

CCS Concepts: • **Human-centered computing** → **Interactive systems and tools**; *Empirical studies in HCI*.

Additional Key Words and Phrases: Gaze Behavior Analysis, Query-Driven Interaction, Human-AI Interaction, Visual Attention, Interactive Systems

1 Introduction

Understanding human gaze behavior is central to behavioral assessment and developmental research [24, 34, 39], as patterns of visual attention provide key signals about social engagement, joint attention, and cognitive processes [26]. In naturalistic interaction settings—such as face-to-face communication, play-based observation, or behavioral studies—observers must simultaneously engage in the interaction while monitoring subtle gaze shifts across people and objects. However, reliably capturing and interpreting such behavior remains challenging due to fundamental limitations in existing workflows.

Current approaches fall into two broad categories. Instrumented gaze sensing, such as wearable or screen-based eye trackers, provides precise measurements but often constrains natural interaction or introduces compliance issues, particularly for young participants [44]. Alternatively, video-based observation preserves ecological validity but relies heavily on time-consuming post-hoc annotation [17]. These workflows create a critical bottleneck: while behavior unfolds dynamically during interaction, meaningful analysis is delayed, labor-intensive, and difficult to scale.

Beyond data collection, there is also a huge gap between direct gaze point prediction and behavioral interpretation. Modern gaze estimation models [12, 50, 54] can predict gaze points or heatmaps with increasing accuracy, yet these low-level outputs do not directly align with how practitioners reason about behavior. In practice, observers ask questions such as “*Did the child look at the other person?*”, “*How long did the participant attend to a specific object?*”, or “*Did attention shift after a cue?*”—queries that require grounding gaze into semantic entities [13, 33, 60] and aggregating events over time.

Finally, existing systems [61] provide limited support for flexible, on-demand behavior analysis within real-world workflows. Researchers and practitioners often need to ask context-specific questions after or during an interaction, rather than following a fixed analysis pipeline. However, traditional tools require manual inspection of timelines or predefined metrics [28], making it difficult to efficiently access targeted behavioral evidence.

To address these challenges, we present **GazeAgent**, a system for *query-driven analysis of gaze behavior* in naturalistic interactions. Instead of treating gaze analysis as a fixed pipeline or a purely perceptual task, GazeAgent reframes it as a structured reasoning problem: transforming raw interaction data into a queryable representation that supports flexible gaze behavioral questions.

GazeAgent follows an observer-centered sensing paradigm, where interaction data is captured unobtrusively during real-world sessions and processed into a structured representation of gaze behavior. The system integrates a multi-stage perception pipeline that combines promptable gaze estimation [12] with scene understanding and grounding modules to associate gaze signals with semantically meaningful entities. These signals are aligned over time to construct a structured *gaze timeline*, consisting of discrete events such as object fixations, eye contact, and gaze shifts.

On top of this representation, GazeAgent introduces a query-driven analysis framework powered by a code-generating agent [5]. Given a natural language query [40, 52], the system translates the query into executable logic over the structured timeline, enabling precise computation of behavioral statistics, temporal relationships, and event sequences. This design moves beyond end-to-end generation by ensuring that outputs are derived from verifiable intermediate representations, supporting accurate and interpretable behavioral analysis.

We evaluate GazeAgent through both technical benchmarks and user studies. Results show that the system achieves robust end-to-end reasoning performance across diverse query types and maintains high consistency in real-world scenarios. In user studies with participants performing gaze analysis tasks, GazeAgent significantly reduces perceived workload and improves task efficiency, enabling faster and more confident interpretation of gaze behavior.

Our contributions are as follows:

- **A query-driven paradigm for gaze behavior analysis.** We reconceptualize gaze analysis from a fixed, perception-centric pipeline into a query-driven reasoning problem, enabling flexible, natural-language access to gaze-based behavioral information in naturalistic interactions. This paradigm bridges the gap between low-level gaze estimation and high-level behavioral interpretation required in research and assessment workflows.
- **A gaze-centric representation with semantic grounding and temporal structure.** We introduce a unified pipeline that transforms raw interaction data into a structured gaze timeline by integrating promptable gaze estimation with scene-aware semantic grounding. This representation encodes gaze behavior as discrete, temporally aligned events (e.g., fixations, eye contact, gaze shifts), providing a computation-ready abstraction for downstream analysis.
- **A verifiable analysis framework via code-generating agents.** We design a query execution framework in which natural-language questions are translated into executable programs over structured gaze data. This approach enables precise computation of temporal, statistical, and relational properties while avoiding hallucination-prone end-to-end generation, ensuring interpretability and reliability in behavioral analysis.
- **Empirical evidence of improved efficiency and usability in gaze analysis.** Through technical evaluation and user studies, we demonstrate that GazeAgent supports robust end-to-end reasoning across diverse query types and significantly reduces analysis workload while improving efficiency and user confidence in interpreting complex gaze behaviors.

2 Related Work

2.1 Interactive Systems for Behavioral Assessment in Clinical Settings

Existing systems for assessing gaze and social behavior in autism and related clinical settings can be broadly grouped into instrumented gaze sensing, retrospective video-based observation, and AR-based interactive technologies. Traditional eye-tracking approaches can provide fine-grained behavioral measurements, but they often require children either to engage in constrained screen-based tasks or to wear sensing hardware. This is a poor match for natural face-to-face clinical interaction, where behavior unfolds through unconstrained engagement with real social partners rather than controlled stimuli [17]. Moreover, although head-mounted eye tracking can be feasible in autistic children, it may also introduce substantial compliance and comfort burden. Perkovich et al. [44] report frequent challenging behaviors during head-mounted eye-tracking sessions, including device removal, elopement, and distress, especially among younger autistic children. These findings suggest that child-worn sensing is not only a measurement choice, but also an interactional intervention that may alter the very behavior clinicians aim to observe [17, 44].

To preserve a more natural interactional setting, many developmental and clinical workflows instead rely on video recording followed by retrospective annotation. ELAN is a widely used framework for multimodal audio-video annotation and was explicitly developed for time-accurate manual coding in multimodality research [62]. However, such workflows remain labor-intensive and shift analysis to after the interaction rather than supporting low-barrier access to behavioral evidence within the clinical workflow. This limitation is also reflected in autism-related sensing work. Chong et al. [17] note that manual video annotation is difficult to scale for screening and outcome assessment, while Alvari et al. [3] emphasize that real-world behavioral analysis in therapist-child interaction remains challenging because the setting is highly dynamic, signals can be interrupted or degraded, and more structured setups often improve technical performance only at the cost of flexibility and ecological value. Thus, fixed-camera or post-hoc video workflows better preserve non-invasiveness than child-worn sensing, but they still impose substantial downstream annotation and interpretation burden [3, 17, 62].

A separate line of work has explored augmented and virtual reality in autism, but this literature has been oriented primarily toward intervention rather than clinician-centered assessment. Berenguer et al.'s systematic review examines the impact of augmented reality in children and adolescents with autism across social, cognitive, and behavioral domains, framing AR mainly as a treatment or support technology [8]. Similarly, Dechsling et al.'s scoping review surveys VR/AR studies for social-skills interventions in autism and shows that immersive technologies have mainly been used to train or improve social functioning rather than to support observation and review in real clinical encounters [22]. These studies are highly relevant because they show where the field has concentrated its design effort: AR/VR has largely been used to deliver intervention, not to help clinicians capture, inspect, and query behavior during or immediately around face-to-face assessment workflows [8, 22].

Overall, prior work reveals a clear interaction gap. Existing approaches either prioritize precise measurement in constrained setups, preserve ecological validity through video capture but defer interpretation to labor-intensive post-hoc analysis, or use AR/VR primarily for intervention delivery. What remains underexplored is a clinician-centered interaction paradigm that combines non-intrusive capture, a socially central viewpoint, and low-barrier access to behavioral evidence. GazeAgent is motivated by this gap: rather than instrumenting the child or requiring extensive retrospective coding, it uses clinician-worn egocentric capture and supports flexible querying of gaze behavior within the clinical workflow [3, 17].

2.2 Gaze Estimation and Visual Scene Understanding

Human gaze target estimation has transitioned from predicting static pixel locations in images [46] to modeling time-varying attention targets in dynamic video sequences [19]. Recent benchmarks, such as Gaze On Objects (GOO), have further shifted the task toward identifying specific object-level bounding boxes rather than just spatial coordinates [59]. Modern architectures have significantly advanced this field by incorporating visual foundation models and transformer-based reasoning. For instance, ViTGaze [54] and Sharingan [57] leverage interaction features and multi-person modeling, while Gaze-LLE [50] demonstrates the power of adapting frozen, large-scale encoders like DINOv2 for low-level gaze localization. Together, these models provide a robust foundation for identifying where a subject's attention is directed within a complex visual field.

Despite these advancements, traditional gaze estimation models often rely on explicit intermediate inputs, such as pre-defined head bounding boxes, which presents a significant bottleneck in clinical environments. In settings involving pediatric assessments, children frequently move quickly, turn away from the camera, or become partially occluded by clinicians and objects. If an upstream detector fails to localize the head, the entire gaze reasoning pipeline often collapses. Recent work on promptable gaze target estimation by Cao et al. [12] addresses this by allowing the subject to be specified through text or visual prompts. This flexibility is vital for GazeAgent, as it enables clinicians to initiate gaze tracking through natural descriptions—such as "the child in the red shirt"—rather than requiring manual bounding box initialization or relying on fragile automated person detectors.

However, gaze localization alone is insufficient for clinical assessment; the resulting coordinates must be grounded in the surrounding physical and social scene to be interpretable. Open-vocabulary detectors like OWL-ViT [38], Grounding DINO [33], and Detic [65] provide the necessary tools to identify arbitrary objects—such as specific toys or clinical tools—from free-form text. To achieve the spatial precision required to determine if a gaze point truly intersects an object, promptable segmentation models like SAM [29] and SAM 3 [13] provide pixel-level masks that surpass the resolution of standard bounding boxes. GazeAgent integrates these perceptual primitives into an agentic workflow, bridging the gap between low-level visual heatmaps and the high-level, temporal behavioral semantics required to answer complex questions about a child's social engagement and developmental milestones.

2.3 LLM Agents and Natural-Language Interfaces

Natural-language interfaces have long been explored as a way to reduce the barrier to data analysis and visualization. Early visualization systems such as Eviza enabled users to ask natural-language questions about existing visualizations [52], while NL4DV translated natural-language queries into analytic and visualization specifications [40]. More recent LLM-based systems further reduce the gap between high-level analytic intent and executable data operations. LIDA uses LLMs to summarize datasets, propose visualization goals, generate visualization code, and produce data-faithful visual outputs [23]. Data Formulator introduces concept-driven visualization authoring, where users define high-level data concepts in natural language and an AI agent performs the necessary data transformation and visualization generation [61]. These systems show the value of using natural language to make data analysis more accessible to non-expert users.

LLM agents extend this idea from query translation to tool orchestration and code execution. InstructPipe allows users to prototype machine-learning pipelines through text instructions by using LLM modules and a code interpreter to generate and render visual programming pipelines [66]. Commercial agentic coding systems such as Claude Code similarly demonstrate how language models can inspect codebases, edit files, run commands, and coordinate multi-step software workflows [5]. These systems are relevant to GazeAgent because gaze behavior analysis requires more than a single model call: the system must parse the clinician's question, retrieve the relevant time interval, select appropriate visual and gaze tools, compute temporal statistics, generate visualizations, and summarize results in natural language.

In parallel, AR and smart-glass agents have begun to explore proactive assistance in the physical world. AiGet integrates AR smart glasses with large multimodal models to deliver context-aware incidental knowledge during everyday activities [11]. Sensible Agent studies how proactive AR agents can adapt both what assistance to provide and how to deliver it using real-time multimodal context, including gaze, hand availability, and environmental noise [31]. These systems demonstrate the potential of AR agents for situated, context-aware assistance, but they are not designed for clinical behavioral assessment or for querying structured gaze timelines from clinician-worn recordings.

GazeAgent connects these threads. Unlike desktop natural-language data analysis systems, it operates over egocentric, spatiotemporal clinical interaction data captured in the physical world. Unlike general AR assistants, it is specialized for clinician-centered behavioral assessment. By combining an LLM code agent with gaze estimation, scene grounding, and visualization tools, GazeAgent enables clinicians to ask open-ended questions about children's gaze behavior and receive interpretable temporal answers without manually constructing analysis pipelines.

3 Gaze Assessment Context and Design Rationale

To ensure that GazeAgent addresses genuine practical bottlenecks rather than artificial technical problems, our system design is grounded in real-world settings where gaze behavior is observed and interpreted. Gaze information serves as a key behavioral signal for understanding social attention in both psychological assessment and developmental research for kids [43]. However, reliably capturing and analyzing kid's gaze behavior in naturalistic interaction remains challenging due to both methodological and practical limitations [10, 41, 53].

In this section, we first review the role of gaze behavior in psychological, developmental, and clinical contexts, as well as the limitations of existing measurement workflows. Based on these observations, we then derive four design guidelines (DGs) that inform the hardware configuration, perception pipeline, and interaction paradigm of GazeAgent.

3.1 Gaze as a Behavioral Signal

Gaze behavior serves as a foundational indicator of social and cognitive development, making its observation central to both clinical practice and psychological research. In clinical workflows, such as the Autism Diagnostic Observation Schedule (ADOS-2) [34], practitioners evaluate social communication through structured yet naturalistic face-to-face interactions. Similarly, developmental frameworks like the Early Social-Communication Scales (ESCS) [39] and the Brief Observation of Social Communication Change (BOSCC) [24] require assessors to meticulously record eye contact, shared attention, and gaze shifts between people and objects.

Beyond immediate diagnosis, gaze patterns offer deep insights into broader social functioning, including attention allocation and responsiveness to environmental stimuli [26, 37]. For instance, extensive research highlights that individuals with Autism Spectrum Disorder (ASD) often allocate significantly less attention to social stimuli than their typically developing peers—a divergence that becomes increasingly pronounced in dynamic, socially rich environments [16, 27].

Despite its critical value, capturing and analyzing gaze behavior reliably during real-world interactions remains a significant hurdle. The necessity of manual coding and qualitative observation introduces a persistent tension between ecological validity (studying behavior in natural settings) and analytical scalability (the ability to process large volumes of high-resolution data).

3.2 Challenges in Real-World Assessment

3.2.1 The Ecological Validity Gap Conventional eye-tracking paradigms often rely on screen-based setups or child-worn sensing devices. While these approaches can provide high-precision measurements, they also alter the setting being observed. On one hand, screen-based paradigms do not reflect how children behave during spontaneous, face-to-face interactions with real social partners [17, 26]. On the other hand, requiring young children—especially autistic children who often present sensory sensitivities—to wear head-mounted eye trackers can introduce substantial practical and behavioral challenges. Perkovich et al. [44] report that among autistic children, 88.8% of 1–4-year-olds and 29.4% of 5–8-year-olds exhibited at least one challenging behavior during head-mounted eye-tracking sessions, with common issues including device removal, elopement, and distress. Such challenges can reduce the naturalism of the interaction and complicate accurate assessment. This limitation also extends to developmental research, where controlled laboratory paradigms may fail to capture how attention is allocated during spontaneous, real-world interactions.

3.2.2 The Assessment–Analysis Disconnect To preserve a more naturalistic environment, both clinical assessment and developmental research often rely on external video recordings of social interaction, followed by offline manual annotation using tools such as ELAN [48]. While this avoids intrusive instrumentation, the process is highly labor-intensive: one validated protocol found that coding five minutes of parent–child interaction video required approximately 60 minutes of trained clinician time [7].

Moreover, this post-hoc workflow provides little support during the live session itself. As a result, clinicians must rely largely on real-time observation and judgment while simultaneously guiding the interaction and tracking brief, easily missed gaze behaviors [1, 36, 37]. This creates a disconnect between the assessment process, which unfolds in real time, and the analysis process, which is delayed and resource-intensive.

A similar challenge arises in developmental research. Large-scale analysis of gaze behavior often depends on manual annotation or simplified proxies, which limits scalability and makes it difficult to systematically capture temporal patterns of attention across extended interactions. Consequently, both clinicians and researchers face challenges in obtaining timely, consistent, and interpretable measurements of gaze behavior in naturalistic settings.

3.2.3 *The Semantic Gap in Gaze Interpretation* Beyond challenges in data collection and analysis, there is a fundamental limitation in how gaze behavior is represented and interpreted. Existing gaze estimation approaches primarily output low-level signals such as gaze points, fixation locations, or heatmaps [19, 56]. While these representations are useful as perceptual primitives, they do not directly capture the behavioral meaning of gaze.

In both clinical assessment and developmental research, the interpretation of gaze depends critically on its target. For example, clinicians are not only concerned with where gaze falls in image coordinates, but whether the child is attending to a specific person or object, how long such attention is sustained, and how it evolves over time [19, 56]. This view is consistent with prior clinical and developmental studies, which analyze gaze behavior in relation to attention toward objects and social stimuli, and emphasize its role in social cognition and learning [16, 27, 34]. As a result, raw gaze coordinates alone are insufficient for supporting behaviorally meaningful analysis.

3.3 Interaction Logic: The Observer-Centric Paradigm

Before detailing our specific design goals, it is essential to define the interaction model of *GazeAgent*. Our system is centered around a human **observer**—who may be a clinical psychologist, a developmental researcher, or a trained practitioner—wearing AR smart glasses.

In this paradigm, the AR device functions as a mobile, multimodal workstation. We define two primary operational roles for the observer: (1) **Active Participant**: The observer directly engages in social interaction with the child, where the glasses capture a first-person view of the engagement. (2) **Side Observer**: The observer monitors the interaction from a vantage point without directly participating, providing a specialized perspective on the dyadic exchange.

Regardless of the specific role, the system treats the captured data as an *interaction-centric* video stream. The observer uses natural language to query behavioral insights, and the system provides feedback through the AR display or synchronized devices. This flexible definition ensures that *GazeAgent* remains applicable across diverse settings, from formal diagnostic clinics to naturalistic laboratory-based behavioral studies.

3.4 Design Guidelines for GazeAgent

To bridge the ecological validity gap and reduce the assessment–analysis disconnect identified above, we derive five core design guidelines. Together, these guidelines motivate the sensing configuration, scene-understanding pipeline, and interaction design of *GazeAgent*.

3.4.1 DG1: Minimize Child-Side Burden via Observer-Worn Sensing A fundamental prerequisite for evaluating social communication and developmental behaviors is the preservation of a naturalistic, unconstrained context. Spontaneous gaze behavior should reflect the individual’s internal attention rather than a reaction to intrusive equipment. Previous works with head-mounted eye trackers or body-worn sensors can not resolve this issue [44].

However, simply relocating sensors to the environment (e.g., fixed tripod cameras) often fails to capture the dynamic nuances of social exchanges, as children frequently move, turn, and shift their focus beyond the field of a stationary lens [53]. Furthermore, visible environmental recording setups can act as distractors, potentially biasing the behavioral data.

These constraints motivate an *observer-worn* sensing paradigm. By embedding multimodal capture into the smart glasses worn by the observer, the system follows the interaction from a socially relevant viewpoint without any instrumentation on the child. This approach maximizes ecological validity across both clinical assessments and psychological research, ensuring that the sensing process remains unobtrusive to the subject while remaining mobile and adaptable to the fluid nature of human interaction.

3.4.2 *DG2: Ground gaze into semantically meaningful scene entities* To support behaviorally meaningful interpretation, gaze-analysis systems should transcend raw spatial coordinates and further relate gaze to observable entities within the environment. This process involves mapping gaze data onto semantic targets, such as specific people or objects and aggregating these grounded observations over time to characterize patterns of attention. For instance, rather than tracking pixel-wised (x, y) coordinates, the system should identify repeated fixations on a specific toy or sustained engagement with a social partner's face.

Such grounding is vital in naturalistic settings, where attention unfolds dynamically and must be interpreted relative to the contextual scene rather than isolated spatial locations [9, 21, 34, 42, 49, 64]. For mobile deployments, this necessitates scene-aware representations that maintain consistent associations between gaze and environmental entities across frequent viewpoint changes, utilizing methods such as persistent object tracking or spatially stable environmental mappings.

3.4.3 *DG3: Facilitate On-Demand, Question-Driven Behavioral Analysis* Information needs in psychological and developmental research are often emergent and contextual, arising dynamically during or after an interaction rather than following a rigid, predefined sequence [2, 20]. To support this workflow, gaze-analysis systems should proactively address context-specific needs, allowing researchers to seek insights without adapting to passive interaction paradigms [35].

However, providing these insights must not interfere with the clinician's primary task: maintaining active social engagement with the child. Continuously presenting analytics in a clinician's field of view can increase mental workload and disrupt the natural flow of interaction [47]. Therefore, a robust system must decouple the capture process from the analysis process. While video is recorded continuously in the background, feedback should be provided strictly on-demand—either during natural pauses or during post-interaction review.

This decoupling enables a shift from exhaustive manual exploration of dashboards and timelines—which are often inefficient for answering spontaneous questions [28, 30]—to flexible, question-driven access. In practice, this is best facilitated through natural language interfaces. By allowing users to pose specific contextual queries and receive task-oriented summaries (e.g., "Did the child look at the toy after I pointed?"), the system provides behavioral disambiguation only when explicitly requested, ensuring the clinician maintains full situational awareness during the live session [51].

3.4.4 *DG4: Ensure Verifiable and Hallucination-Free Reasoning for Clinical Trust* While natural language interfaces provide the flexibility required for on-demand analysis (DG3), deploying generative AI in clinical settings necessitates stringent reliability. Traditional end-to-end Vision-Language Models (VLMs) operate as "black boxes" and are notoriously prone to hallucinations, particularly when tasked with precise mathematical counting, temporal sequencing, or tracking fine-grained behavioral shifts over long videos. In a behavioral assessment context, providing an incorrect count of eye contact events or hallucinating a non-existent gaze shift can actively misguide clinical judgment.

Therefore, the system architecture must prioritize verifiability and transparency over purely neural generation. Rather than directly generating textual answers from video inputs, the underlying AI should operate as an agentic planner that translates the clinician's natural language query into transparent, executable logic (e.g., Python scripts). By executing these programmable queries over the structured, semantically grounded session metadata (established in DG2), the system can compute exact statistics, durations, and temporal correlations with absolute mathematical correctness. This symbolic reasoning approach ensures that every extracted insight—whether a simple frequency count or a complex cross-event timeline—is fundamentally hallucination-free, thereby establishing the crucial clinical trust needed for real-world adoption.

3.4.5 *DG5: Ensure Data Privacy and Clinical Compliance* The sensitive nature of pediatric clinical assessments imposes strict ethical and legal constraints on data handling, such as HIPAA compliance and Institutional

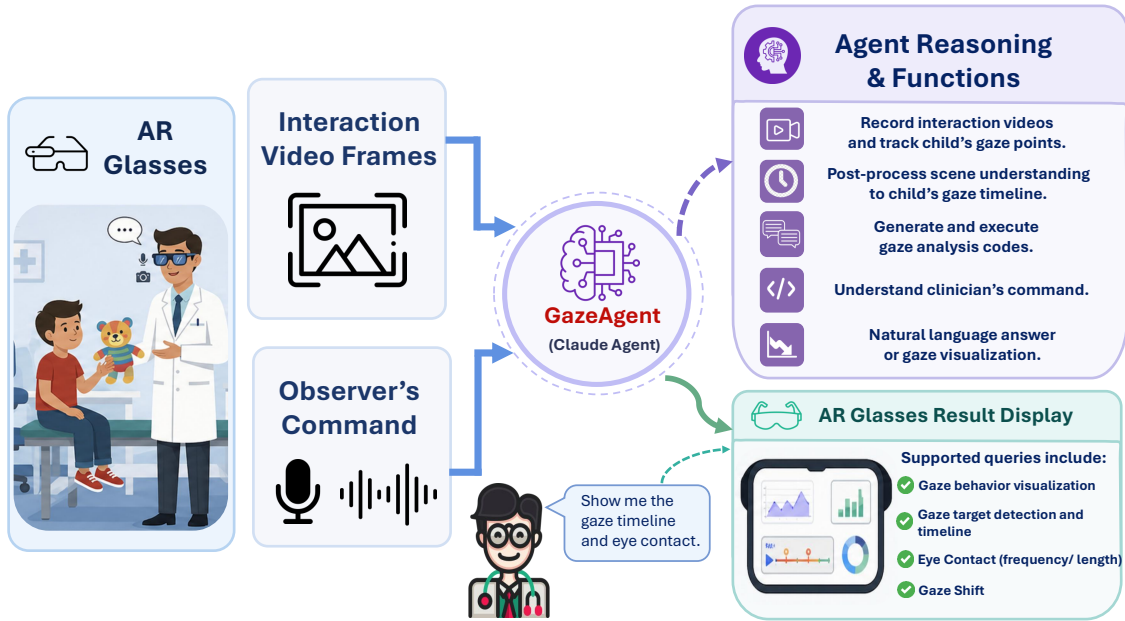


Fig. 2. The interaction loop of *GazeAgent*. Unobtrusive AR glasses capture multimodal inputs (video and audio), while the central Claude Agent dynamically translates the clinician’s natural language queries into executable reasoning codes for real-time gaze analysis. (Part of the figure is created by Generative AI)

Review Board (IRB) regulations. Clinical interactions inherently capture highly sensitive, personally identifiable information, including children’s faces, distinct behaviors, and confidential medical conversations. Relying on prevalent cloud-based multimodal APIs (e.g., GPT-4o) would require transmitting unanonymized video streams to external third-party servers. This introduces unacceptable privacy risks and violates standard clinical data protection protocols. Therefore, the system design must fundamentally prioritize data sovereignty, ensuring that all observational data is processed securely without leaving the clinical environment.

4 GazeAgent System Architectural Design

We introduce *GazeAgent*, an end-to-end wearable system designed to facilitate non-intrusive, naturalistic observation and automated analysis of pediatric gaze behaviors. As illustrated in Figure 3, the system architecture is deliberately designed to fulfill the design guidelines (DGs) established in Section 3.4. It bridges the gap between unstructured egocentric video and actionable clinical insights through a continuous data flow comprising three primary modules: (1) Interaction Video Capture, (2) Gaze-Centric Perception and Semantic Grounding, and (3) Query-Driven Analysis via a Code Agent.

To contextualize the *GazeAgent* architecture, we describe a typical interaction flow involving a clinician (or psychologist) and a child:

- **Initialization:** At the start of a behavioral assessment, the clinician equips the smart glasses and initiates the system via a descriptive voice command (e.g., “Start recording, track the child in the blue shirt”). This initial utterance serves a critical dual purpose: it triggers the camera to begin the monitoring session and simultaneously sets the target subject description. This textual description acts as the foundational

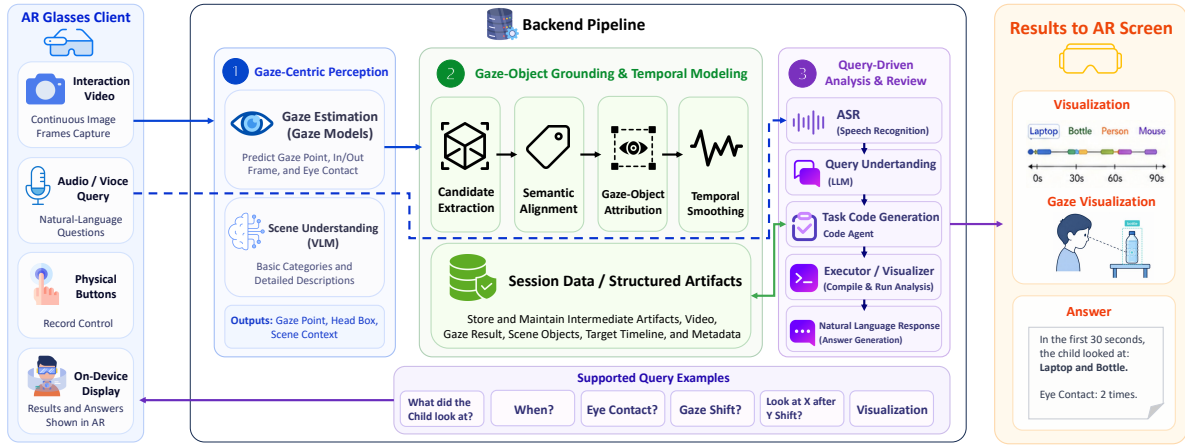


Fig. 3. System architecture of GazeAgent. The system captures video (Module 1), processes it through a scene-aware perception pipeline using GazeAnywhere [12], Qwen 3.5 [58], SAM3 [13] and SigLIP2 [60] to ground gaze into semantic entities (Module 2), and employs a Claude-driven Code Agent to dynamically execute natural language queries and generate verifiable visualizations (Module 3). (Part of the figure is created by Generative AI)

prompt for the *GazeAnywhere* module, enabling the model to precisely isolate and lock onto the correct individual in the dynamic environment from the very first frame.

- **Unobtrusive Observation:** During the interaction, GazeAgent operates in the background to preserve ecological validity. The system processes captured frames through a pipeline of foundation models that identify the child, social partners, and environmental objects. These observations are serialized into a behavioral metadata stream, documenting the child's gaze target sequences and object interactions in real-time.
- **Session Conclusion:** At the conclusion of the assessment, the clinician manually terminates the recording via a simple voice command or device interaction. Upon this deliberate closure, the system immediately finalizes the structured *Session Data Artifacts*, officially transitioning the workflow from the background capture phase to the interactive query phase.
- **Interactive Querying:** The clinician then transition to an analytical mode, posing natural language questions such as: "How much eye contact occurred between me and the child?"
- **Visual Feedback:** GazeAgent parses the query against the generated metadata to provide a precise quantitative answer and renders supporting visualizations directly onto the smart glass display for immediate behavior review.

4.1 Module 1: Interaction Capture and Recording Pipeline via Smart Glass (Addressing DG1)

To preserve the naturalistic flow of the assessment interaction without instrumenting the child (DG1), *GazeAgent* utilizes an smart glasses client as the primary sensory frontend. During the live session, the observer wears the glasses, which continuously record video capturing the child and the surrounding environment from a socially central, first-person perspective. Crucially, to prevent cognitive overload during the live interaction (DG3), the glasses act purely as a passive data collection and audio-query interface in this phase. The raw video stream is transmitted to the backend pipeline, decoupling the heavy computational analysis from the clinician's immediate visual field.

4.2 Module 2: Gaze-Centric Perception and Grounding (Addressing DG2)

The core challenge in processing clinical video is moving beyond raw 2D gaze coordinates to understand *what* the child is looking at over time. The backend perception pipeline achieves this semantic grounding (DG2) by orchestrating several state-of-the-art vision models to construct a structured Gaze Timeline.

4.2.1 Subject Localization and Gaze Estimation For robust gaze tracking without manual calibration, we utilize **GazeAnywhere** [12], a promptable gaze target estimation foundation model. Given a natural language prompt (e.g., “the child”), GazeAnywhere pinpoints the subject and generates three primary outputs: (1) the subject’s bounding box, (2) the predicted gaze point in pixel-wised (x, y) , and (3) an *In/Out-of-frame* indicator.

The *In/Out-of-frame* flag serves as a critical gating mechanism for behavioral disambiguation. When the child is looking at an object within the camera’s field of view (“In”), the system utilizes the 2D gaze point for semantic object grounding. However, a significant portion of social communication in children social behavior assessment—specifically *eye contact* with the observer—occurs when the subject looks directly at the wearer (the observer), which often results in an “Out-of-frame” signal. To verify whether such instances constitute genuine social engagement rather than random gaze shifts, we integrate a specialized **Eye-Contact CNN** [18]. This model specifically analyzes the subject’s facial features to detect eye contact. By combining GazeAnywhere’s spatial tracking with the Eye-Contact CNN’s social detection, *GazeAgent* can accurately log gaze events both within the physical environment and across the social dimension.

4.2.2 Hierarchical Scene Understanding: Qwen3.5, SAM 3, and SigLIP2 Given the computational asymmetry between high-frequency gaze tracking and heavy vision-language inference, our scene-understanding pipeline operates asynchronously. A locally deployed **Qwen 3.5-27B** [58] processes video key frames at a 2-second interval to catalog environmental objects. To overcome the known limitations of text-driven segmentation models in grounding highly complex or lengthy descriptions, we engineered Qwen to generate hierarchical labels: broad categories (e.g., “toy”) and detailed semantic descriptions (e.g., “red toy car,” “blue toy dinosaur”).

We then employ a coarse-to-fine grounding strategy. First, the broad category labels are passed to **SAM 3 (Segment Anything Model 3)** [13] operating in video mode. SAM 3 reliably detects these general objects, extracting pixel-wise masks that are subsequently converted into stable bounding boxes. Next, to associate the detailed semantics with the correct spatial locations, we introduce **SigLIP2** [60] as a zero-shot vision-language matching bridge. For each bounding box identified by SAM 3 under a specific category, SigLIP2 computes similarity scores against all corresponding detailed descriptions. These descriptions are then exclusively assigned to the highest-scoring bounding boxes, effectively resolving multi-object ambiguity within the same category.

4.2.3 Semantic Alignment and Temporal Smoothing Following the hierarchical scene understanding, the *Gaze-Object Attribution* module aligns the real-time gaze signals with the semantic environment. It calculates the spatial intersection between the high-frequency gaze points (continuously generated by GazeAnywhere) and the semantically enriched bounding boxes (updated every 2 seconds). To handle momentary tracking noise caused by rapid head movements or transient occlusions, a *Temporal Smoothing* algorithm filters the intersection data over short rolling windows. The final output of this pipeline is a structured, temporally aligned database (Session Data Artifacts) containing discrete, high-fidelity gaze events (e.g., [00:15 - 00:18] Child looks at red toy car).

4.3 Module 3: Query-Driven Analysis via Code Agent (Addressing DG3)

Rather than forcing clinicians to manually scrutinize the generated timelines, *GazeAgent* introduces a multi-tool agentic architecture to support flexible, event-driven summarization (DG3).

To ensure a seamless transition from clinical observation to interactive analysis, *GazeAgent* employs an **asynchronous parallel processing** strategy. While the clinician is recording the session, the *GazeAnywhere* and **Qwen 3.5** models run continuously in the background. By the time the recording ends, the foundational

gaze coordinates and high-level scene contexts are already indexed in the *Session Data Artifacts*. Consequently, the heavy computational load is significantly front-loaded; only the initial query requires a one-time execution of **SAM 3** and **SigLIP2** to finalize precise spatial grounding. This design minimizes the clinician’s waiting time during the subsequent interactive query phase.

4.3.1 Audio Transcript Conversion To bridge natural verbal interactions and backend computational processes, *GazeAgent* incorporates a streaming speech-to-text pipeline. We employ the **faster-whisper** [45, 55](*large-v3*) model to convert spoken queries into machine-readable text. Audio is captured client-side and streamed to the server via WebSockets, where it is dynamically segmented into discrete utterances. To ensure high transcription accuracy for specialized terminology, the buffered audio is submitted to Whisper alongside a domain-specific *initial prompt* listing gaze-tracking vocabulary (e.g., *fixation*, *saccade*, *gaze trajectory*). This strategically biases the decoder toward the controlled command grammar of our system. Finally, the decoded segments are concatenated into a single transcript and forwarded to the Claude-driven Code Agent for intent parsing and code generation.

4.3.2 Claude-Driven Code Agent (Central Dispatcher) Vibe coding based on **Claude** serves as the brain of the system. When the user provides a natural language query, the *Query Understanding* module (powered by **Claude Sonnet 4.6**) [6] parses the linguistic intent. Instead of relying on a static QA model, the Code Agent dynamically generates and executes Python scripts to query the structured *Session Data Artifacts*. This programmatic approach—orchestrated by **Claude Haiku 4.5** [4] for rapid sub-task execution—ensures that statistical counts, temporal sequences, and event correlations are computed with absolute logical accuracy.

4.3.3 Taxonomy of Supported Analytical Queries Because the Code Agent operates on a structured, temporally aligned database rather than raw pixels, it supports a highly expressive query taxonomy. Crucially, **all queries support precise temporal slicing** (e.g., “*in the first 30 seconds*,” “*from 0:30 to 1:30*”), allowing clinicians to isolate specific interaction phases. The system natively supports four categories of clinical tasks:

- **Entity-Centric Queries (Objects & Eye Contact):** Clinicians can query specific behavioral targets, including environmental objects and social partners (i.e., eye contact with the interactor). The Code Agent supports Boolean checks (“*Did the child look at the car?*”), temporal localization (“*When?*”), and duration accumulation (“*For how long?*”).
- **Macro-Behavioral Profiling:** To support holistic assessment, the Agent can generate a complete *Gaze Shift Timeline*, summarizing the chronological sequence of attention allocation throughout a specified period.
- **Sequential and Relational Logic:** The most complex, yet clinically vital, queries involve temporal dependencies. The Agent can programmatically resolve sequential conditions (e.g., “*What did the child look at **after** I shifted the toy?*” or “*Did they make eye contact **before** looking at the book?*”).

4.3.4 On-Demand Visualization and Verifiability Once the Code Agent computes the answer, it synthesizes a concise Natural Language Response. To mitigate automation bias and ensure clinical verifiability (DG4), the system pairs text answers with on-demand visual evidence. The Executor automatically retrieves the exact video clips corresponding to the query and overlays non-intrusive visualizations—such as **dynamic gaze points and object bounding boxes**—onto the playback. Additionally, for macro-level queries, the system can render a **static Gaze Timeline visualization** (e.g., a Gantt chart of gaze events). These results are presented back to the clinician’s interface on-demand, allowing them to rapidly cross-check the AI’s conclusions against the augmented behavioral footage without wading through the entire recording.

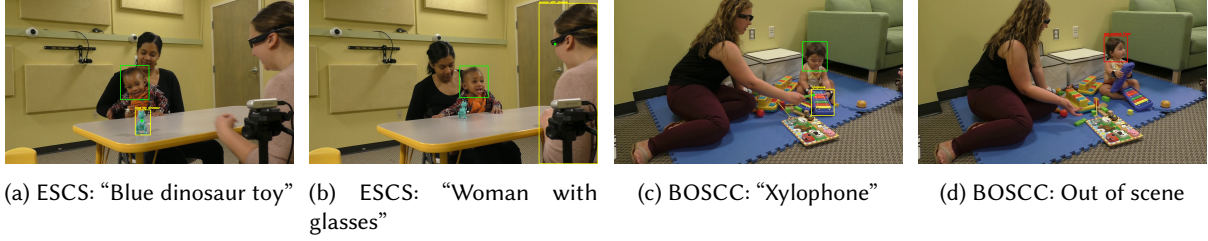


Fig. 4. Qualitative examples of gaze target grounding across different clinical evaluation protocols (ESCS and BOSCC). **Annotations:** The green dot represents the child's estimated gaze point; green and red bounding boxes indicate the tracked interaction subjects; and yellow bounding boxes demarcate the localized target objects. These examples demonstrate the system's robust capability in **open-vocabulary object grounding**. By processing complex, free-form linguistic descriptions (e.g., "blue dinosaur toy," "woman with glasses," "xylophone"), the underlying vision-language model successfully achieves fine-grained visual-semantic alignment, precisely locating unstructured objects within cluttered clinical environments or accurately predicting out-of-scene attention shifts.

4.4 Implementation and Technical Stack

The *GazeAgent* backend is implemented as a modular Python service, adopting a "privacy-first" architecture to directly fulfill the data sovereignty requirements of **DG5**. To protect patient identity and clinical confidentiality, we implement a strict **informational boundary** between raw visual perception and high-level reasoning.

On-Premise Perception Layer. All computer vision (CV) modules involving raw pixel processing are deployed exclusively on local workstations, ensuring that identifiable visual data never leaves the clinical environment. Specifically, the system utilizes an NVIDIA RTX 6000 Pro GPU to host the **Qwen 3.5-27B** for semantic summarization. A secondary dual-GPU workstation handles spatial and behavioral grounding: an NVIDIA RTX 3080 runs **GazeAnywhere** and our custom **Eye-Contact CNN**, while an NVIDIA RTX 4090 is dedicated to the high-memory requirements of **SAM 3** and **SigLIP 2**.

Isolated Reasoning via Cloud-Local Decoupling. For high-level linguistic reasoning and Python code generation, *GazeAgent* leverages Claude 4.6 Sonnet and Claude 4.5 Haiku via secure APIs. Crucially, these cloud-based models operate within a **strict visual sandbox**: they are explicitly denied access to any image-based content, video streams, or raw sensor data. Instead, our architecture decouples perception from reasoning by converting local visual outputs into **symbolic, non-identifiable metadata** (e.g., JSON-based object lists and gaze coordinates). The Claude agents interact exclusively with these structured *Session Data Artifacts*, performing logical analysis and code execution on abstract tokens rather than visual signals. This decoupling ensures that the computationally intensive "Perception" remains local and private, while the "Reasoning" leverages advanced LLMs without risking a breach of HIPAA or institutional privacy protocols.

5 Technical Evaluation

To rigorously validate the robustness and clinical consistency of *GazeAgent* prior to our user study, we conducted a targeted technical evaluation. Rather than benchmarking isolated visual algorithms, this evaluation assesses the system's end-to-end capability to translate natural language into executable spatial-temporal reasoning scripts within noisy, real-world clinical environments, similar in spirit to execution-based evaluation in code generation and semantic parsing, where outputs are assessed based on functional correctness rather than surface-form similarity [15, 63].

Query Category	Sample (N)	Execution %	Consistency %	Primary Metric
Entity & State	20	100%	95.0% (19/20)	Exact Match
Quantitative	30	100%	83.3% (25/30)	Clinical Tolerance
Temporal-Relational	40	100%	80.0% (32/40)	Hybrid Tolerance
Boundary Defense	10	N/A	100% (10/10)	Rejection/Clarification
Total / Average	100	100%	86.0%	–

Table 1. Performance Summary of *GazeAgent* across 100 Naturalistic Queries. The evaluation categorizes queries by logical complexity, applying type-specific metrics to ensure clinical consistency.

5.1 Experimental Setup and Metrics

We selected five representative interaction video clips (ranging from 1 to 2 minutes each) captured during face-to-face clinical interactions. To establish a robust ground truth (GT), we manually annotated these videos frame-by-frame. Prior to annotation, all coders underwent a structured training session led by a clinical domain expert to establish a standardized protocol for identifying and delimiting key gaze events.

Query Generation Protocol: To evaluate the system within its intended interactive context, we authored a test bank of 100 queries (20 per video) and deliberately avoided randomized query generation in favor of simulating naturalistic practitioner inquiries, reflecting task-oriented evaluation settings commonly used in interactive AI systems. Consequently, these queries focus on clinically salient events—such as primary toys being manipulated and mutual gaze dynamics—rather than exhaustively probing every background object. The queries span four categories of increasing logical complexity.

Mixed-Tolerance Evaluation Framework: Given the non-deterministic nature of LLM generation and the inherent tracking noise in dynamic clinical videos, we implemented a type-specific evaluation framework inspired by prior work that distinguishes between logical correctness and robustness under noisy or perturbed inputs [14, 15]. This design penalizes logical hallucinations while accommodating acceptable physical tracking margins:

- **Entity & State (N=20):** Queries identifying objects or verifying boolean states (e.g., “What toys did the child look at?”). These are evaluated using a *Strict Exact Match* criterion (zero tolerance for omissions or semantic hallucinations), consistent with exact-match evaluation used in semantic parsing benchmarks [63].
- **Quantitative (N=30):** Statistical queries (e.g., “How many times did the child shift their gaze?”). These are evaluated using a *Clinical Tolerance* margin aligned with human inter-rater reliability (± 1 absolute error for $GT < 5$; $\pm 20\%$ margin for $GT \geq 5$), reflecting acceptable variability observed in behavioral coding and annotation agreement [25].
- **Temporal-Relational (N=40):** Complex sequential and conditional logic queries (e.g., “Did the child look at the laptop after the block?”), consistent with video question answering settings where answers must be grounded in temporally localized evidence [32]. These are evaluated using a *Hybrid Metric*: Strict Exact Match for sequence verification, and a strict ± 1.0 -second window for temporal grounding.
- **Boundary Defense (N=10):** Ambiguous or out-of-scope queries designed to test system guardrails (e.g., “Was the child engaged?”). Success strictly requires the agent to explicitly request clarification or refuse unsupported cognitive inferences.

5.2 Results and Error Analysis

As detailed in Table 1, *GazeAgent* achieved a 100% code execution success rate and an overall clinical consistency of 86.0%. Notably, the system demonstrated flawless boundary defense (100%), successfully prompting users to specify ambiguous targets and strictly refusing to make unsupported abstract cognitive inferences.

An in-depth analysis of the 14 consistency failures reveals a critical system insight: the current limitations of our pipeline lie primarily in the foundational *perceptual grounding layer* rather than the Agent’s reasoning logic. We identified three main sources of perceptual cascading errors:

- (1) **Tracking Instability in Segmentation Model (SAM3):** The segmentation model occasionally struggled with the dynamic physical occlusions common in child-clinician interactions. We observed two distinct manifestations of this limitation. First, *Occlusion-Induced Fragmentation*: when small toys were manipulated and partially obscured by the subject’s hands, SAM3’s tracking would intermittently drop out depending on the viewing angle. The object mask would temporarily vanish and reappear as the physical occlusion shifted. This fragmentation of a single continuous interaction into disjointed tracked segments caused false-positive inflations in *Quantitative Queries* (e.g., overcounting gaze shifts) and *Temporal-Relational Queries*. Second, *Delayed Initialization*: in scenarios where an object entered the frame at an extreme angle or under partial occlusion, SAM3 experienced a delayed onset in establishing a stable track. This truncation directly resulted in underestimations in gaze duration queries, accounting for specific failures in the *Temporal-Relational* category.
- (2) **Spatial Resolution and Pose Ambiguity in Gaze Estimation:** While the gaze estimation model robustly handled general interactions, challenging viewing angles—such as extreme downward head tilts during focused play—occasionally introduced slight spatial offsets in the predicted gaze vectors. In highly cluttered scenes where multiple interactive objects were clustered together, these small trajectory variances caused the predicted gaze point to inadvertently cross the bounding box from the actual target (Object A) to an adjacent object (Object B). These artificial crossovers may introduce false-positive gaze shifts in *Quantitative Queries* and directly disrupt the strict chronological logic evaluated in *Temporal-Relational* queries. Consequently, this leads to sequence misclassifications (e.g., incorrectly logging a gaze shift between toys when the subject’s true focus remained uninterrupted on a single item).
- (3) **Semantic Omissions in Vision-Language Grounding (VLM):** A small subset of failures highlighted a critical reliance on the VLM’s comprehensive initial scene parsing. In rare instances, the VLM failed to explicitly label specific toys in its initial text description, even when those objects were later actively engaged with. This created a “semantic vacuum”: although SAM3 and the gaze model correctly tracked the spatial coordinates of the physical interaction, the Code Agent lacked the specific vocabulary to query the object by name. This initial grounding gap inherently bypassed our downstream logic, resulting in inevitable failures in the corresponding *Entity & State* queries, as well as cascading omissions in any subsequent sequential queries involving those missing targets.

Overall, these findings confirm that *GazeAgent* provides a highly robust logical reasoning and execution framework. By systematically decomposing the errors, we demonstrate that the current bottlenecks are isolated within the physical perception and initial semantic mapping layers. As these foundational visual models continue to evolve, the end-to-end clinical consistency of our pipeline will naturally scale without requiring modifications to the Agent’s core architecture.

6 User Study 1: Controlled Experiment on Cognitive Offloading

To comprehensively evaluate the utility, usability, and integration potential of *GazeAgent* as a collaborative clinical copilot, we designed a two-part mixed-methods evaluation. In this section, we present **Study 1**, a controlled

within-subjects experiment focusing on cognitive load reduction and visual observation strategies among novice clinical annotators.

Evaluation Scope and Rationale. While *GazeAgent* is designed as an end-to-end wearable system (encompassing both AR capture and backend reasoning), conducting an in-situ clinical deployment with real autistic children wearing experimental AR hardware presents severe ethical, logistical, and privacy challenges.

To rigorously and safely evaluate our core technical contribution—the LLM-powered gaze reasoning and visual annotation backend—we adopted a modular evaluation strategy. In both Study 1 and Study 2, we isolated the backend Agent and evaluated its utility using pre-recorded clinical videos. This post-hoc review paradigm directly mirrors the system’s intended "on-demand" analytical workflow (DG4) and represents the most intensive use case for clinical behavioral analysis. By holding the video perspective constant, we could strictly isolate and quantify how the Agent’s analytical outputs impact clinician workload and decision-making.

6.1 Experiment Design

Participants. We recruited 16 participants (12 female, 4 male) who were compensated for their time. To ensure the ecological validity of the evaluation, we enforced strict pre-screening criteria: all participants were required to have completed relevant academic coursework in child development and possess in-person interaction experience with children. This rigorous selection guarantees that our cohort represents "novice annotators" or clinical trainees—users who hold foundational domain knowledge to interpret gaze behaviors but lack the ingrained, rapid intuition of senior clinicians.

Materials and Tasks. The study materials comprised four 1-minute video clips sourced from standard clinical assessment protocols: two from the Early Social Communication Scales (ESCS) and two from the Brief Observation of Social Communication Change (BOSCC). For each video, participants completed a visual search and interpretation task by answering three multiple-choice questions designed to simulate rapid clinical triaging: (1) identifying the overall gaze pattern, (2) estimating the frequency category of gaze shifts, and (3) determining contingency (whether the child changed gaze targets following a cue).

Experimental Design and Hypotheses. We employed a within-subjects design with the **Assistance Condition** as the primary Independent Variable, consisting of two levels: *Baseline* (raw video) and *With GazeAgent*. We formulated three primary hypotheses:

- **H1 (Cognitive Offloading):** The use of *GazeAgent* will significantly reduce the overall subjective workload, particularly in the *Mental Demand* and *Effort* dimensions.
- **H2 (Task Efficiency):** Participants will report higher confidence and perceived performance when answering gaze-related questions with AI assistance.
- **H3 (System Utility):** The visual annotations will be rated as highly effective for disambiguating complex social interactions.

To rigorously mitigate learning and order effects, the 16 participants were evenly assigned to 8 distinct groups (2 participants per group) using a Latin Square counterbalancing matrix. This matrix dictated a sequence of four trials for each participant, ensuring that the presentation order of the video protocols (ESCS vs. BOSCC) and the alternating assistance conditions (*With* vs. *Without*) were perfectly counterbalanced across the study. The complete 8-group counterbalancing matrix is detailed in Appendix A.

Procedure. Each session lasted approximately 30-40 minutes and strictly followed a standardized protocol:

- (1) **Introduction:** We verified participants’ backgrounds and introduced the study design, task objectives, and the NASA-TLX items to ensure complete clarity.

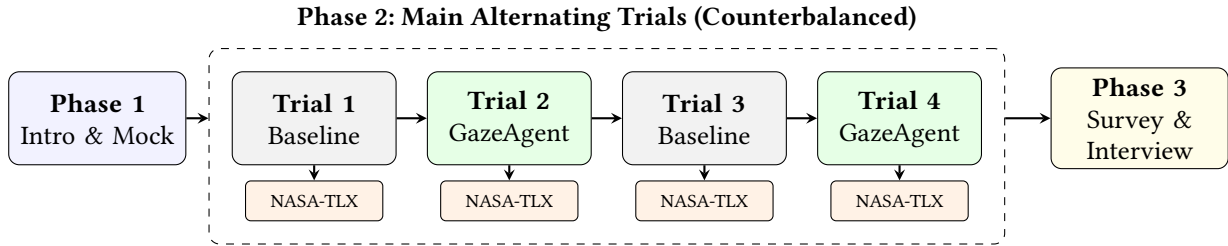


Fig. 5. Experimental procedure for a single participant in Study 1. The main session consisted of four sequential trials. The starting condition and video protocols were perfectly counterbalanced using a Latin Square matrix across 8 groups (detailed in Appendix A). NASA-TLX was administered immediately after each trial.

- (2) **Familiarization Phase:** Participants completed a full mock trial using an irrelevant, non-protocol video to acclimate to the workflow and eliminate learning effects before the formal study.
- (3) **Main Trials:** Participants completed four sequential trials based on their assigned group. The trials strictly alternated between the *Baseline* and *With GazeAgent* conditions. In the *Baseline* condition, participants watched the raw video, optionally re-watched or scrutinized specific segments, and then answered the clinical questions. In the *With GazeAgent* condition, participants watched the video and actively queried the Agent for analytical results (visual annotations and textual reasoning). They could then verify these insights before finalizing their answers. Crucially, immediately after each of the four trials, participants completed the NASA-TLX to capture their workload state in real-time.
- (4) **Post-Study Assessment:** Upon finishing all four trials, participants answered two Likert-scale questions regarding their overall experience. Finally, we conducted a brief semi-structured exit interview to capture the qualitative rationale behind their NASA-TLX ratings. The interview focused on three main directions derived from real-world clinical observation needs: (1) how the Agent's visual outputs assisted in identifying specific gaze behaviors across varying complexities, (2) how the integration of the Agent altered their traditional observation workflow and visual attention, and (3) their perception of the Agent's limitations (e.g., tracking instability) and its impact on their analytical confidence.

Data Analysis Strategy. Recognizing the inherent individual variability in subjective baseline ratings (NASA-TLX), our analysis focused on the within-subject differences. Thanks to the strict counterbalancing of the ESCS and BOSCC protocols, we first aggregated the data by calculating the average score for each participant under both the *Baseline* and *With GazeAgent* conditions.

For the seven dependent variables (the Overall Weighted TLX and its 6 sub-dimensions), we computed the paired difference ($\Delta = \text{Baseline} - \text{GazeAgent}$). We then conducted Paired-samples t-tests to evaluate whether the mean differences were significantly distinct from zero. To ensure statistical robustness given our sample size ($N = 16$), we concurrently applied the non-parametric Wilcoxon signed-rank test. Finally, we calculated the paired Cohen's d to determine the standardized effect sizes.

6.2 Quantitative Results: Workload Reduction and Task Utility

Our quantitative analysis of the NASA-TLX and post-study survey provides strong statistical evidence supporting the cognitive benefits of *GazeAgent*. Consistent with our data analysis strategy, we reported both parametric (Paired-samples t-test) and non-parametric (Wilcoxon signed-rank test) results to ensure robustness, alongside Cohen's d for effect sizes.

6.2.1 Quantitative Results: Workload Reduction and Task Utility
 Given our sample size ($N = 16$), we first conducted a Shapiro-Wilk test on the paired differences to assess data normality. The test revealed that the Overall Weighted TLX differences significantly deviated from a normal distribution ($W = 0.835, p = .008$). Consequently, to maintain statistical rigor across all measurements, we abandoned parametric tests and exclusively utilized the Wilcoxon signed-rank test. Accordingly, we report the Median and Interquartile Range (IQR [Q1, Q3]), which directly correspond to the distributions shown in our box plots.

Overall Workload (H1): The analysis revealed a statistically significant reduction in the overall weighted NASA-TLX score when participants used *GazeAgent* (Median = 35.63, IQR = [28.20, 48.97]) compared to the raw video Baseline (Median = 46.72, IQR = [28.48, 56.13]). The Wilcoxon signed-rank test confirmed this substantial decrease ($W = 10.0, p = .001^{**}$). This strongly supports **H1** and demonstrates that the AI copilot fundamentally offloads the overall cognitive burden required for clinical observation tasks (see Figure 6).

Individual NASA-TLX Dimensions: To understand the specific sources of this workload reduction, we examined the six sub-dimensions of the NASA-TLX (see Figure 7). The data revealed that the overall reduction was primarily driven by significant decreases in two critical cognitive dimensions:

- **Mental Demand [*]:** Participants reported significantly lower mental exertion with the Agent (Median = 40.00, IQR = [27.50, 60.00]) compared to the Baseline (Median = 52.50, IQR = [34.38, 70.00]); $W = 9.5, p = .012$. This highlights the system's effectiveness in reducing the perceptual arithmetic of tracking rapid eye movements.
- **Effort [*]:** The perceived effort required to achieve their level of performance was also significantly reduced in the *GazeAgent* condition (Median = 50.00, IQR = [26.88, 58.75]) versus the Baseline (Median = 53.75, IQR = [31.88, 70.63]); $W = 25.5, p = .027$.

Additionally, we observed a marginally significant improvement in perceived **Performance** (where lower scores indicate greater perceived success and less failure), with the Agent condition scoring a Median of 28.75 (IQR = [17.50, 40.63]) versus 36.25 (IQR = [14.38, 56.88]) in the Baseline; $W = 22.0, p = .100$. The remaining dimensions (Physical Demand, Temporal Demand, and Frustration) yielded no statistically significant differences ($p \geq .38$). The lack of significance in Temporal Demand is particularly logical, as the video lengths (1 minute) and the core analytical tasks remained temporally constant across both conditions.

Task Utility and User Ratings (H2 & H3): Beyond the standardized workload metrics, the post-study descriptive statistics reinforced the system's practical value. When asked to rate how effectively the system helped them complete the clinical observation tasks, participants awarded *GazeAgent* a high mean score of 4.12 out of 5 ($SD = 0.72$). This strong positive consensus aligns with **H2** and **H3**, quantitatively validating that the integration

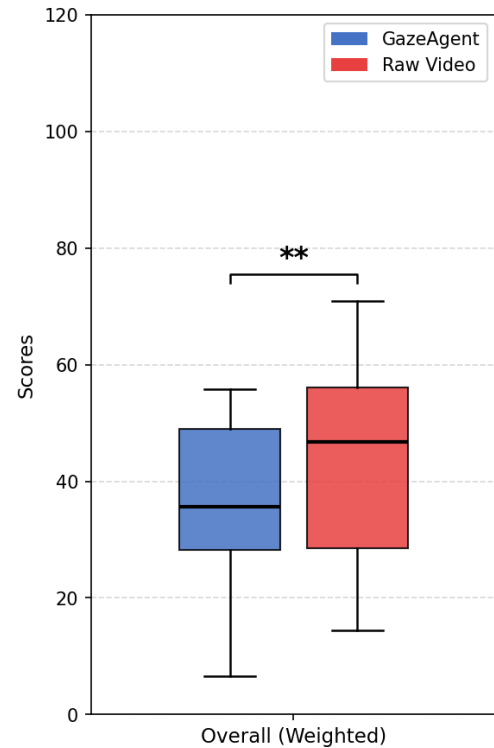


Fig. 6. Overall NASA-TLX weighted scores comparing the Baseline and With *GazeAgent* conditions. The Agent significantly reduced the overall workload (** $p < 0.01$).

of visual annotations and AI-driven insights serves as a highly useful mechanism for analyzing complex social interactions.

6.3 Qualitative Insights and Thematic Analysis

By triangulating the post-study interview feedback, we identified six core themes that deeply contextualize the quantitative results. To systematically understand how *GazeAgent* influences clinical observation, we categorized these themes into three dimensions: the mechanisms of assistance (Pros), the required shift in observation strategy (Neutral), and the fundamental algorithmic and clinical limitations (Cons).

1. Enhancing Observation Accuracy through Spatial Salience (Pro).

In traditional clinical observation, annotators constantly balance cognitive workload against observation accuracy. Subtle behavioral changes are easily overlooked, and achieving high accuracy typically demands exponential mental effort. *GazeAgent* disrupts this trade-off by utilizing object bounding boxes and gaze dots as visual highlighters for micro-behaviors. Multiple participants noted that the system made their observations fundamentally more accurate because changes were immediately salient. As P9, P13, and P16 explicitly highlighted, in complex scenarios where a child makes "frequent, small gaze shifts," the changes are "hardly distinguishable in raw video." The AI's spatial annotations act as an instant visual cue, ensuring that these fleeting but clinically significant attention shifts are captured rather than missed, thereby intrinsically linking workload reduction with higher diagnostic precision.

2. Validating Judgments and Reducing Psychological Uncertainty (Pro).

Beyond pure discovery, the Agent provided crucial assistance through validation. In raw video analysis, uncertainty often forces annotators to frequently rewind and re-watch segments, significantly increasing temporal and mental demand. Participants utilized the Agent as a collaborative "double-checker" (P1) to validate their own initial interpretations. The trajectory visualizations provided "more insight into the exact start and end points of a gaze shift event" (P12, P14, P15), cementing their temporal judgments. Furthermore, in highly ambiguous edge cases—such as determining whether a child is looking at an object or staring blankly outside the scene—the system's "in/out of frame prediction" served as a decisive reference (P2). This mechanism successfully "avoided the uncertainty that usually requires rewatching the video" (P1, P13), translating into higher subjective confidence.

3. Longitudinal Workflow Integration and Rapid Video Retrieval (Pro).

While the immediate workload reduction for a single, one-off video review might appear subtle, participants with psychological research backgrounds (P3, P14, P16) recognized the system's profound value for longitudinal workflows. Research videos are rarely used just once; they are continuously archived, re-evaluated, and coded over time. For this overarching process, P3, and P14 noted that the system's auto-generated summaries can significantly streamline repetitive analysis, with P3 explicitly referring to the outputs as a valuable "pre-coding." Building on this, P14 and P16 emphasized that during the study tasks, having a timeline distribution allowed them to "quickly grasp the overall state and distribution of the child's attention" before diving into tedious frame-by-frame inspection. Extrapolating this immediate experience to their broader research backgrounds, they envisioned that this macro-level visualization could effectively serve as a rapid retrieval index. Therefore, assuming continuous improvements in detection accuracy, what feels like a modest assist in a single-session trial promises to save substantial effort in the iterative, long-term analysis typical of behavioral research.

4. The "Split-Attention" Dilemma and the Tool Learning Curve (Neutral).

Despite the benefits, introducing an AI copilot into a well-established clinical workflow is not a frictionless, plug-and-play experience. It necessitates a fundamental paradigm shift in how annotators visually scan the environment. Participants reported that adding more UI elements created a "split-attention" effect: "Before, I only

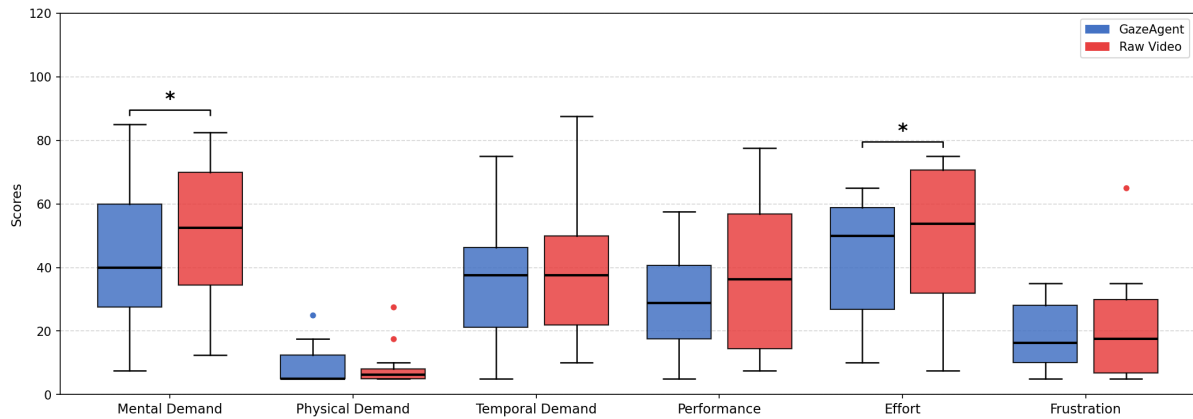


Fig. 7. Boxplots detailing the six NASA-TLX sub-dimensions across conditions. Significant reductions were observed in Mental Demand (* $p < 0.05$) and Effort (* $p < 0.05$). No statistically significant differences were found in Physical Demand, Temporal Demand, Performance, or Frustration ($p \geq 0.05$).

focused on the children, but then I also needed to focus on the annotations" (P15). This indicates that the tool does not automatically reduce effort; rather, it replaces the effort of "guessing gaze directions" with the effort of "integrating AI visual outputs." Consequently, this new observation strategy inherently possesses a learning curve. P6 and P7 emphasized that adapting to the annotations requires dedicated practice. P6 noted that "even after one mock trial, I wasn't really well-versed in this tracking," suggesting that novice annotators need sufficient training time specifically to build new visual habits for human-AI collaborative observation.

5. Trust Disruption via Algorithmic Jitter (Limitation).

The acceptance of the system heavily depended on how it handled ambiguity. Participants demonstrated tolerance for minor, stable AI misclassifications (P10), but their trust was severely penalized by mechanical instability. In difficult cases where a child's face was partially obscured, the AI's gaze point would exhibit "frequent shaking" or rapidly "shift between toys" (P4, P16). This erratic behavior causes severe cognitive dissonance. Human annotators possess implicit logical knowledge: even if they cannot see the exact target, they know the child's gaze is steadily following something. The AI's jitter "introduces confusion as it obviously conflicts with human thought" (P16), creating visual noise that ultimately forces the annotator to ignore the Agent and rely solely on the raw video.

6. The Semantic Gap Between Visual Tracking and Clinical Intent (Limitation).

Finally, P5 articulated a profound limitation regarding the boundaries of current computer vision models in psychology. While *GazeAgent* excels as a geometric tracker—mapping the physical trajectory of eyes to objects—true clinical observation is fundamentally about social communication. The clinical value lies not just in *where* the child looks, but *why* they are looking (e.g., requesting, sharing interest, or avoidance). P5 argued that understanding gaze behaviors "should reflect intent, and the annotation didn't help in understanding the intent part." This highlights a critical ceiling for the system: while it effectively offloads the perceptual burden of physical tracking, the deep semantic reasoning regarding psychological intent remains an exclusively human responsibility.

7 Study 2: Ecological Validation with Frontline Clinicians

While Study 1 demonstrated the system’s quantitative benefits for novice annotators., clinical practice ultimately relies on the expertise of senior professionals. To evaluate the ecological validity, clinical utility, and real-world integration potential of *GazeAgent*, we conducted Study 2—an in-depth, qualitative system validation with frontline clinicians.

7.1 Methodology

Participants. As frontline clinicians are often very busy, we purposefully sampled 4 senior clinicians (e.g., developmental pediatricians, clinical psychologists) with extensive clinical experience ranging from 8 to 13 years (Mean = 11.5 years). Unlike the novice cohort in Study 1, these experts possess years of ingrained diagnostic intuition and represent the ultimate end-users of clinical AI copilots. Their participation was completely voluntary.

Materials and Apparatus. To strictly test ecological validity, we deliberately avoided the standardized protocol videos used in Study 1. Instead, the evaluation utilized a set of real-world clinical consultation videos provided by an independent group of doctors. These unconstrained videos capture the true complexity of clinical settings, including unstructured interactions and cluttered backgrounds. For each scenario, we presented both the raw footage (Baseline) and the *GazeAgent*-processed outputs.

Procedure. Each session lasted approximately 30 minutes, conducted individually with the experts. The procedure consisted of three distinct phases:

- (1) **System Onboarding (10 min):** We first introduced the design rationale and core capabilities of *GazeAgent*, including a demonstration of its natural language query interface and how the system processes gaze behaviors.
- (2) **Comparative Video Review (10 min):** Clinicians were presented with the unconstrained clinical videos. They carefully reviewed and compared the raw, unannotated footage (Baseline) against the *GazeAgent*-processed outputs. This phase allowed the experts to independently evaluate the system’s performance in handling real-world visual noise and complex interactions.
- (3) **In-depth Evaluative Interview (10 min):** Immediately following the video review, we conducted a semi-structured interview to capture their expert evaluations. The discussion focused on three macro-level dimensions: (1) *Clinical Validity* (how well the AI’s observations align with professional clinical judgments), (2) *Workflow Integration* (how such a system might augment or alter current diagnostic pipelines), and (3) *System Boundaries* (limitations, trust issues, and reliability concerns).

Data Analysis. The experts’ feedback and our detailed interview notes were aggregated and analyzed using a thematic approach. We iteratively reviewed the qualitative data to extract and group the core insights, ultimately yielding four high-level themes that illuminate the complex dynamics of human-AI collaboration in real-world clinical settings.

7.2 Results: Qualitative Feedback from Experts

The thematic analysis of the expert interviews revealed four core dimensions that highlight both the transformative potential of *GazeAgent* and the practical challenges of deploying AI in high-stakes clinical workflows.

1. Filtering Visual Noise in Unconstrained Clinical Environments (Pro).

Real-world clinical consultations are inherently messy. As P1, P3, and P4 noted, these unconstrained environments feature multiple social targets (e.g., parents, other medical staff) and cluttered backgrounds filled with toys, office supplies, and medical equipment. In raw videos, tracking a child’s gaze through this visual noise imposes a severe

cognitive load and introduces constant distractions. The experts highly praised *GazeAgent* for effectively cutting through this clutter. By predicting and explicitly visualizing gaze targets, the system acts as an attentional filter. It allows clinicians to easily follow the child's gaze behavior, shielding them from environmental distractions and helping them "focus more intently on hard-to-observe segments" (P1). Furthermore, P2 highlighted that this filtering mechanism not only aids micro-level tracking but also provides a "quick, overarching understanding of the interaction process."

2. Paradigm Shift: Corroborating Subjective Intuition with Objective Data (Pro).

A profound insight emerged regarding how *GazeAgent* could fundamentally shift clinical assessment methodologies. Currently, evaluations for conditions like autism heavily rely on the clinician's subjective feelings and intuitive judgments formed during live interactions. P1 envisioned that the system's data-driven summaries could serve as an objective anchor to "corroborate the doctor's subjective intuition." Rather than replacing the clinician's gut feeling, the AI provides a quantifiable backbone. Both P1 and P4 strongly validated the system's clinical utility, noting that this hybrid approach—merging human intuition with empirical behavioral data—can significantly elevate diagnostic efficiency and confidence.

3. NL Accessibility and the Need for Event-Driven Alignment (Constructive Feedback).

The experts (P1, P4) strongly approved of the natural language (NL) querying interface, noting that it significantly lowers the adoption barrier for medical staff without technical backgrounds. However, P1 provided a critical design critique: querying based purely on timestamps is cognitively unnatural for clinicians. Doctors often struggle to perceive and recall the exact temporal boundaries of fleeting micro-behaviors. Instead, P1 and P4 strongly advocated for shifting from time-driven to *event-driven* alignment. They suggested that the AI's output should align gaze behaviors with specific external social stimuli or the doctor's own interactive actions (e.g., calling the child's name). Mapping AI observations directly to these interaction milestones would seamlessly match the doctor's subjective experience, making it much easier to assess crucial clinical metrics, such as distinguishing between voluntary and involuntary social interactions.

4. Automation Bias and the Necessity of Clinical Calibration (Limitation).

Echoing the findings from Study 1, the experts confirmed that introducing AI overlays disrupts existing visual scanning patterns. P2 and P3 noted that clinicians would require structured training and adaptation periods before applying the system in actual diagnoses. More importantly, P2 and P4 raised serious concerns regarding "automation bias"—the psychological tendency for humans to over-rely on automated outputs. Given the critical nature of clinical diagnoses, P2 warned that doctors might "blindly depend on erroneous AI predictions" if they trust the system too easily. Therefore, P4 stressed that while the system demonstrates immense clinical potential, it mandates comprehensive stress-testing across diverse, edge-case clinical conditions before deployment. Establishing strict trust boundaries and understanding the model's reliability limits are as critical as the algorithmic accuracy itself.

8 Discussion

8.1 From Gaze Estimation to Queryable Analysis

A key contribution of *GazeAgent* is the shift from treating gaze estimation as a standalone perception task to supporting gaze as a queryable analytical representation. Traditional gaze systems focus on predicting low-level outputs such as gaze points or heatmaps, which require substantial manual interpretation before they can support meaningful analysis. In contrast, our approach introduces an intermediate abstraction—a structured gaze timeline—that encodes gaze behavior as temporally aligned, semantically grounded events.

This representation enables a different mode of interaction with gaze data. Rather than inspecting raw predictions or manually constructing metrics, users can directly query behavior through natural language. This reframes gaze analysis from a perception-centric pipeline into a reasoning-oriented process, where the system supports answering questions about attention, duration, and temporal relationships. More broadly, this suggests that the value of gaze estimation systems may lie not only in improving prediction accuracy, but in how their outputs are structured and made accessible for downstream analysis.

8.2 Query-Driven Access to Behavioral Evidence

Our findings highlight the importance of query-driven interaction as an alternative to traditional analysis workflows. Existing approaches, including manual annotation and dashboard-based exploration, require users to either exhaustively label data or navigate predefined metrics. These approaches assume that analysis goals are known in advance, which is often not the case in real-world behavioral observation.

In contrast, GazeAgent supports on-demand access to behavioral evidence through natural-language queries. This allows users to dynamically specify what information they need, when they need it, without committing to a fixed analysis pipeline. In our user study, participants reported improved efficiency and confidence, suggesting that this interaction model better aligns with how behavioral questions arise in practice. More generally, this points to a shift from “exploring data” toward “asking targeted questions,” which may be broadly applicable to other forms of observational data analysis.

8.3 Reliability Through Verifiable Reasoning

Another important aspect of our design is the use of structured representations and executable logic to support verifiable reasoning. End-to-end generative approaches, while flexible, are often prone to hallucination and inconsistencies when performing temporal counting or relational reasoning over long sequences. In contrast, GazeAgent separates perception from reasoning: perceptual models produce structured outputs, and a code-generating agent operates over this data to compute results deterministically.

This design improves interpretability and reliability, as outputs can be traced back to explicit intermediate representations. At the same time, our technical evaluation shows that system errors are primarily driven by limitations in the perception pipeline—such as tracking instability, spatial ambiguity, or incomplete scene grounding—rather than failures in the reasoning process itself. This suggests that while structured reasoning can provide a robust analytical layer, overall system performance remains bounded by the quality of underlying perceptual signals.

8.4 Limitations

Despite these contributions, GazeAgent has several limitations. First, the system is fundamentally dependent on the quality of its perception components. Errors in gaze estimation, object grounding, or temporal alignment can propagate to downstream analysis, leading to incorrect or incomplete results. As observed in our evaluation, challenges such as occlusion, closely spaced objects, and missing semantic labels can affect the accuracy of the constructed gaze timeline.

Second, the current system focuses exclusively on gaze-based signals. While gaze is a critical indicator of visual attention, it captures only a subset of observable behavior. The system does not model other modalities such as body movement, gestures, or speech, nor does it attempt to infer higher-level cognitive states or intentions. As a result, the scope of analysis is limited to gaze-related patterns and their temporal structure.

Third, the expressiveness of supported queries is constrained by the underlying representation. Although the system supports temporal, statistical, and relational queries over gaze events, more complex forms of behavioral reasoning—particularly those involving implicit context or abstract interpretation—remain out of scope.

Finally, while GazeAgent enables efficient post-hoc analysis, it does not yet support fully integrated real-time interaction. The current design separates data capture from analysis to reduce cognitive load, but tighter integration between observation and querying may be desirable in some settings.

8.5 Future Directions

Our findings suggest several promising directions for extending query-driven gaze analysis systems.

From single-modality to multi-modal behavioral events. User feedback indicates a need for richer behavioral descriptions beyond gaze alone. Extending the system to incorporate additional modalities—such as body pose, hand movement, or gestures—could enable the construction of more fine-grained event representations (e.g., reaching, pointing, or coordinated gaze-action patterns). However, such extensions depend on robust perception models capable of reliably detecting and aligning these signals in naturalistic environments.

From single-subject to multi-subject interaction analysis. The current system models the gaze behavior of a single target subject. In many real-world scenarios, however, behavior emerges through interactions between multiple participants. Supporting multi-subject analysis—such as jointly modeling gaze, actions, and speech across participants—would enable richer queries about interaction dynamics. Achieving this requires reliable multi-agent tracking and cross-modal alignment, which remain challenging in unconstrained settings.

Toward relational and event-based interaction analysis. Building on these directions, a longer-term opportunity is to support relational queries over interaction events, such as how one participant’s action influences another’s gaze response over time. Such capabilities would allow users to analyze interaction patterns while remaining grounded in observable signals. Realizing this vision requires both richer representations and advances in perception pipelines to support consistent temporal grounding.

Overall, these directions point toward a broader opportunity: extending query-driven analysis from gaze-centric signals to more comprehensive representations of interaction. Importantly, progress in this direction depends not only on system design, but also on the development of perception models that can robustly extract and align behavioral signals in real-world environments.

9 Conclusion

In this work, we present **GazeAgent**, a system for query-driven analysis of gaze behavior in naturalistic interactions. We address key limitations of existing approaches, which either rely on intrusive sensing setups or require labor-intensive post-hoc annotation, and do not directly support how gaze is analyzed in practice.

GazeAgent reframes gaze analysis from a perception-centric task into a structured reasoning process. By transforming raw interaction data into a semantically grounded and temporally aligned gaze representation, and enabling natural-language queries through executable analysis, the system supports flexible access to gaze information without requiring predefined metrics or manual inspection.

Through technical evaluation and user studies, we show that this approach enables robust reasoning over gaze behavior while significantly reducing analysis workload and improving efficiency and user confidence. These results suggest that structuring perceptual outputs into queryable representations can play a critical role in making complex behavioral signals more accessible and actionable.

More broadly, our work highlights the potential of query-driven interaction as a paradigm for analyzing observational data. By bridging gaze perception and interpretable gaze analysis, GazeAgent provides a foundation for future systems that support flexible, on-demand access to structured gaze information in real-world interaction settings.

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A Methodological Transparency and Reproducibility Appendix (META)

A.1 System Implementation and Hyperparameters

To support the reproducibility of the *GazeAgent* system, we detail the underlying model configurations and hyperparameters derived directly from the deployed codebase.

- **Scene-Object VLM (Qwen):** A local Qwen3.5-72B (BF16 native) is deployed via vllm as the primary perception layer. It operates at temperature=0.1 with top_p=0.95. The model uses chunked prefill (max_num_batched_tokens=8192) and FlashAttention v2 to maintain low-latency inference.
- **Claude Code Agent:** The reasoning layer utilizes claude-sonnet-4-6 (temperature=0) for the Formatter and Code Generator. The Naturaliser utilizes claude-haiku-4-5-20251001 to minimize latency.
- **Speech-to-Text (Whisper):** Configured with model="large-v3", compute_type="float16", and beam_size=5. A domain-specific initial prompt containing gaze-tracking vocabulary is injected to enhance clinical transcription accuracy.
- **Spatial Grounding:** SAM 3 state is offloaded to CPU during evaluation to conserve VRAM. SigLIP 2 (so400m-patch14-384) performs phrase-to-crop assignment via Hungarian matching on L2-normalized cosine similarity scores.
- **Gaze and Behavior:** *GazeAnywhere* predictions are binarized at a 0.3 threshold. The *Eye-Contact CNN* uses a 0.9 threshold. Fixation registration requires a minimum of 10 consecutive frames (FIXATION_MIN_FRAMES) and uses a bounding-box expansion tolerance ratio of 0.05.

A.2 Agent Prompting and Reasoning Logic

The system logically decouples visual perception (Qwen) from behavioral reasoning (Claude Code Agent), operating without external orchestration frameworks like LangChain.

Vision-Language Perception (Qwen VLM). Rather than open-ended visual question answering, the local Qwen VLM is tightly constrained to act as a deterministic *strict salient-entity summarizer*.

- **Task Instruction:** The prompt explicitly forbids positional words, actions, or full sentences. It must identify up to 10 salient persons or objects, described strictly by coarse category and detailed appearance.
- **Temporal-Update Mode:** For video analysis, the VLM receives the prior frame's JSON output and emits an updated list. It tracks object permanence by assigning a status of "same", "updated", "new", or "missing", significantly minimizing temporal hallucinations.
- **Robust Parsing:** VLM outputs are forced into JSON formats. A tolerant extractor is implemented to bypass pre/post-amble generative noise.

Semantic Taxonomy Alignment (Claude to SAM 3 / SigLIP 2). To bridge the gap between Qwen's open-vocabulary outputs and the zero-shot grounding modules, Claude performs a critical semantic alignment. Because SAM 3 inherently struggles to align fine-grained descriptive text directly with spatial masks, our system utilizes SigLIP 2. Claude maps raw VLM descriptions into a two-tier hierarchy:

- "human": ["woman in black t-shirt", "boy in white-shirt"]
- "toy": ["red car toy", "blue dinosaur toy"]

This architectural handshake prevents severe bounding-box overlap ambiguities inherent to SigLIP 2 (e.g., a person's bounding box encompassing a held cup, causing the person to be misidentified as the cup). The coarse macro-categories are used for logical deduplication and spatial hierarchy resolution, while the fine-grained vocabulary is fed to the SigLIP 2 text encoder, ensuring highly precise heatmap generation for SAM 3.

Code Agent Reasoning and Execution (Claude). Operating over the grounded metadata, the Claude Agent acts as a three-stage pipeline (Formatter → Code Generator → Naturaliser).

- **Ethical Refusal and Formatting:** Converts utterances into a strict JSON specification. It enforces an **ethical refusal rule**: subjective constructs (e.g., “*Was the child engaged?*”) are mapped to status: “ambiguous” with a concrete reformulation, ensuring the system never hallucinates inner-state measurements.
- **Code Generation:** Emits a single Python function (def run_task(context)). It bypasses explicit Chain-of-Thought (CoT) to reduce latency, instead relying on a curated catalogue of allowed helper functions and 7 few-shot task templates.
- **Result Naturalization:** The Haiku-driven Naturaliser maps the structured execution dictionary into a concise reply using predefined response templates.
- **Defensive Execution Sandbox:** To guarantee safe execution, the host code strips markdown fences and compiles the code using Python’s built-in compile(). Execution occurs via exec() within a strictly isolated sandbox where __builtins__ is emptied to explicitly block eval, import, and open.

A.3 User Study Design and Evaluation Protocol

To ensure empirical transparency, we detail the ethical data handling and experimental logistics. Detailed counterbalancing and participant demographics are provided in Tables 2–4.

Study 1: In-Person Controlled Evaluation. To adhere to strict IRB protocols and prevent unauthorized transmission of sensitive pediatric recordings, Study 1 was conducted exclusively **in-person** using an air-gapped local workstation. Compared to the ESCS, the BOSCC protocol features predefined play scenarios that are inherently more complex and involve a greater number of interactive objects. Because of this inherent disparity, we did not enforce a uniform baseline difficulty across all trials; instead, we strictly matched the observation difficulty *within* each protocol group. Specifically, we first pre-selected interaction clips to balance visual and contextual complexity. Senior clinical experts then independently confirmed the behavioral density of our selections, and subsequent pilot testing empirically validated that videos from the same protocol were equally challenging. Furthermore, trial sequences were perfectly counterbalanced using a Latin Square matrix (see Table 2) to completely mitigate potential order and learning effects.

Study 2: Remote Ecological Validation. Transmitting our private dataset to remotely participating senior clinicians would violate the aforementioned data confinement protocols. To creatively resolve this ethical constraint while evaluating the system “in the wild,” clinicians reviewed unconstrained, real-world consultation videos securely provided from their own practices. This baseline and assisted evaluation was immediately followed by in-depth interviews focusing on clinical validity, workflow integration, and the boundaries of the AI agent.

A.4 Exhibits: Hardware Specifications of the AR Glasses

To rigorously fulfill **DG1 (Minimize Child-Side Burden)** and ensure ecological validity during clinical evaluations, the user interface of *GazeAgent* was deployed on a pair of commercial off-the-shelf augmented reality (AR) glasses RayNeo X3 Pro. The hardware was specifically selected for its socially unobtrusive form factor. Resembling a standard pair of everyday eyeglasses, it completely conceals its computational nature, thereby minimizing visual distraction and social interference when clinicians interact with pediatric patients.

The client application was developed using the Unity engine and compiled for the device’s native Android operating system. Despite its lightweight profile of only 76g, the device is equipped with dual front-facing cameras capable of continuous video recording. It also features built-in microphones and speakers to facilitate seamless voice-based querying. For the visual interface, the glasses utilize a binocular full-color display powered by a Micro LED optical engine, ensuring that clinicians receive clear, on-demand visual feedback without breaking eye contact with the child. Detailed specifications are summarized in Table 5.

A.5 Exhibits: Experimental Design and Materials

Table 2. Counterbalanced Latin Square Design for User Study 1. The 16 participants were evenly assigned to 8 groups. The sequence counterbalances the video stimuli (V1–V4), clinical protocol (E = ESCS, B = BOSCC), and system assistance (W = With GazeAgent, WO = Without GazeAgent).

Group (<i>N</i> = 2 per group)	Trial 1	Trial 2	Trial 3	Trial 4
Group 1	V1 (W) [E]	V2 (WO) [B]	V3 (W) [E]	V4 (WO) [B]
Group 2	V2 (W) [B]	V3 (WO) [E]	V4 (W) [B]	V1 (WO) [E]
Group 3	V3 (W) [E]	V4 (WO) [B]	V1 (W) [E]	V2 (WO) [B]
Group 4	V4 (W) [B]	V1 (WO) [E]	V2 (W) [B]	V3 (WO) [E]
Group 5	V1 (WO) [E]	V2 (W) [B]	V3 (WO) [E]	V4 (W) [B]
Group 6	V2 (WO) [B]	V3 (W) [E]	V4 (WO) [B]	V1 (W) [E]
Group 7	V3 (WO) [E]	V4 (W) [B]	V1 (WO) [E]	V2 (W) [B]
Group 8	V4 (WO) [B]	V1 (W) [E]	V2 (WO) [B]	V3 (W) [E]

Table 3. Participant Demographics for User Study 1 (In-Person Controlled Evaluation, *N* = 16). The cohort comprises undergraduate and graduate students with expertise in psychological and cognitive sciences, and hands-on experience in child behavioral observation.

Participant ID	Gender	Academic Level	Major / Discipline	Experience with Children (Years)
P1	Male	Undergraduate	Clinical Psychology	2.5
P2	Male	Undergraduate	Psychology	2.0
P3	Female	Undergraduate	Clinical Psychology	2.0
P4	Female	Undergraduate	Psychology	<1.0
P5	Female	Graduate (Ph.D.)	Attention and Perception	3.0
P6	Female	Graduate (Ph.D.)	Cognitive Neuroscience	2.0
P7	Male	Graduate (Ph.D.)	Social Psychology	4.0
P8	Female	Graduate (Ph.D.)	Social Psychology	2.5
P9	Female	Undergraduate	Developmental Psychology	2.0
P10	Female	Undergraduate	Psychology & Data Science	4.0
P11	Male	Undergraduate	Developmental Psychology	0.5
P12	Female	Undergraduate	Clinical Psychology	2.5
P13	Female	Undergraduate	Developmental Psychology	3.0
P14	Female	Graduate (M.S.)	Developmental Psychology	3.5
P15	Female	Undergraduate	Psychology	1.0
P16	Female	Graduate (Ph.D.)	Social Psychology	4.5

Table 4. Participant Demographics for User Study 2 (Ecological Validation with Frontline Experts, $N = 4$). All experts are certified professionals actively practicing at a tertiary children’s hospital, with specialized experience in pediatric developmental and behavioral health.

Expert ID	Gender	Clinical Title	Years of Experience
E1	Female	Physiotherapist	13
E2	Male	Pediatrician	8
E3	Male	Occupational Therapist	12
E4	Female	Physiotherapist	9

Table 5. Hardware specifications of the AR Glasses utilized for the *GazeAgent* clinician interface.

Component / Feature	Specification
Operating System	Android (Custom client compiled via Unity)
Form Factor & Weight	Standard eyewear design; 76±1g
Display Solution	Binocular full-color display
Optical Engine	Micro LED Firefly Optical Engine (FOV 30°)
Maximum Brightness	2000 nits (Ensuring legibility in bright clinical rooms)
Vision Sensors	Dual front cameras supporting video recording
Audio Components	Built-in microphones and speakers for voice interaction

Fig. 8. The RayNeo X3 Pro AR glasses serve as the GazeAgent hardware interface. Its 76g lightweight and standard eyewear design minimize social interference while integrating dual cameras and a Micro LED engine for real-time clinical feedback.

